Modes of Convergence and Borel–Cantelli Lemmas

Stats 220A

Lecture overview

Things we do today:

1. Define the four standard modes of convergence for random variables.
2. Prove the key implications and give counterexamples showing the gaps.
3. Prove both Borel–Cantelli lemmas: our first major tool for a.s. arguments.

Why this matters. Almost every theorem in this course is a statement about convergence. Having a clear taxonomy of *what kind* of convergence we mean—and what it costs to prove—is foundational for everything that follows.

Setup

We work throughout on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.
$X, X_1, X_2, \ldots$ are real-valued random variables defined on this space.

We want to formalize "X_n converges to X."

The subtlety. Each X_n is a function $\omega \mapsto X_n(\omega)$, so there are multiple inequivalent notions of "approaching a limit":

- Pointwise for $\mathbb{P}$-almost every ω (almost sure convergence)
- The probability of a large deviation goes to zero (in probability)
- The expected p-th power of the deviation goes to zero (L^p convergence)
- The distributions converge weakly (in distribution)

These are genuinely different notions. We will see all four in this course.

Almost sure convergence

Definition

X_n converges to X **almost surely** ($X_n \xrightarrow{\text{a.s.}} X$) if

$$\mathbb{P}\big(\{\omega \in \Omega : X_n(\omega) \to X(\omega)\}\big) = 1.$$

Equivalently: there exists a null set N with $\mathbb{P}(N) = 0$ such that

$$X_n(\omega) \to X(\omega) \quad \text{for all } \omega \notin N.$$

Intuition. This is ordinary real-analysis pointwise convergence, holding at every ω except on a probability-zero exceptional set. For each individual outcome ω (outside N), the sequence of numbers $X_1(\omega), X_2(\omega), \ldots$ converges to the number $X(\omega)$ in the usual ε-N sense.

This is the strongest among a.s., in prob, and L^p.

Example: almost sure convergence

Let $(\Omega, \mathcal{F}, \mathbb{P}) = \big((0,1],\ \mathcal{B}(0,1],\ \text{Lebesgue}\big)$ and define

$$X_n(\omega) = \omega^n, \qquad \omega \in (0,1].$$

What happens pointwise?

- For $\omega \in (0,1)$: $\omega^n \to 0$ as $n \to \infty$ (geometric decay).
- For $\omega = 1$: $\omega^n = 1$ for every n, so $X_n(1) = 1 \not\to 0$.

Conclusion. Let $N = \{1\}$. Then $\mathbb{P}(N) = 0$, and $X_n(\omega) \to 0$ for every $\omega \notin N$. Therefore

$$X_n \xrightarrow{\text{a.s.}} 0.$$

What this illustrates. The exceptional set N where convergence fails can be non-empty — it just needs to have probability zero. Here the single point $\omega = 1$ causes no trouble because $\mathbb{P}(\{1\}) = 0$.

In this example convergence is actually pointwise on $(0,1)$, so a.s. is not even binding the set. We will see later that a.s. and pointwise are genuinely different once we have more structure.

A useful reformulation of a.s. convergence

Proposition

$X_n \xrightarrow{\text{a.s.}} X$ if and only if for every $\varepsilon > 0$, $\mathbb{P}\big(\sup_{k \geq n} |X_k - X| > \varepsilon\big) \to 0$ as $n \to \infty$.

Proof.

$X_n(\omega) \to X(\omega)$ iff for all $\varepsilon > 0$, eventually $|X_k(\omega) - X(\omega)| \leq \varepsilon$ for all $k \geq n$. In set notation,

$$\{X_n \to X\} = \bigcap_{m=1}^{\infty} \bigcup_{n=1}^{\infty} \bigcap_{k \geq n} \{|X_k - X| \leq \tfrac{1}{m}\}.$$

The sets $B_n^m := \bigcap_{k \geq n} \{|X_k - X| \leq 1/m\}$ increase in n. So $\mathbb{P}(X_n \to X) = 1$ iff $\mathbb{P}(B_n^m) \to 1$ for each m, i.e., $\mathbb{P}\big(\sup_{k \geq n} |X_k - X| > 1/m\big) \to 0$ for each m. $\square$

Key distinction. This asks for *the entire tail* $\{X_k : k \geq n\}$ to stay near X, with probability tending to 1. Compare with in-probability, which only asks about the single term X_n.

Convergence in probability

Definition

X_n converges to X **in probability** ($X_n \xrightarrow{\mathbb{P}} X$) if for every $\varepsilon > 0$,

$$\mathbb{P}(|X_n - X| > \varepsilon) \longrightarrow 0 \quad \text{as } n \to \infty.$$

Head-to-head comparison with a.s.

- *In probability:* at each fixed time n, is X_n likely to be close to X?
- *Almost sure:* do we eventually *stay* close to X, for a.e. ω?

In probability allows the "bad set" $\{|X_n - X| > \varepsilon\}$ to keep *moving* around Ω as long as it gets small in measure. Almost sure convergence forces those bad sets to *eventually disappear* for each ω.

This is why convergence in probability is strictly weaker: the bad set $\{|X_n - X| > \varepsilon\}$ may keep moving with n, as long as its probability goes to 0.

Example: convergence in probability

Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}X_i = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$, and let $\bar{X}_n = n^{-1}\sum_{i=1}^n X_i$.

Claim: $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$.

Proof.

For any $\varepsilon > 0$, Chebyshev's inequality gives
$\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \mathrm{Var}(\bar{X}_n)/\varepsilon^2 = \sigma^2/(n\varepsilon^2) \to 0.$ $\square$

This is the Weak Law of Large Numbers. For any fixed tolerance ε, the probability that $\bar{X}_n$ misses μ becomes negligible. It says nothing about *which* ω's are well-behaved.

Note on what we used. We only needed $\sigma^2 < \infty$. The Strong Law (a.s. convergence) requires $\mathbb{E}|X_i| < \infty$ and substantially more work. This separation motivates why we study both modes carefully.

L^p convergence

Definition

For $p \geq 1$, X_n converges to X **in** L^p ($X_n \xrightarrow{L^p} X$) if $\mathbb{E}|X_n|^p < \infty$, $\mathbb{E}|X|^p < \infty$, and $\mathbb{E}|X_n - X|^p \to 0$ as $n \to \infty$.

Special cases. $p = 1$: convergence in *mean*. $p = 2$: *mean-square convergence*, the most important for statistics.

Proposition (L^p norms are nested)

If $p \geq r \geq 1$ *then* $X_n \xrightarrow{L^p} X$ *implies* $X_n \xrightarrow{L^r} X$.

Proof.

By Jensen applied to the convex function $t \mapsto t^{p/r}$:

$$(\mathbb{E}|X_n - X|^r)^{p/r} \leq \mathbb{E}\left(|X_n - X|^r\right)^{p/r} = \mathbb{E}|X_n - X|^p \to 0. \qquad \square$$

Higher p means stronger mode: $L^p \Rightarrow L^r$ for $p > r$, but not vice versa.

Example: L^2 convergence

Same setup: $X_1, X_2, \ldots$ i.i.d. with $\mathbb{E}X_i = \mu$, $\mathrm{Var}(X_i) = \sigma^2 < \infty$, and $\bar{X}_n = n^{-1}\sum_{i=1}^n X_i$.

Claim: $\bar{X}_n \xrightarrow{L^2} \mu$.

Proof.

$\mathbb{E}\big[(\bar{X}_n - \mu)^2\big] = \mathrm{Var}(\bar{X}_n) = \sigma^2/n \to 0.$ $\square$

Comparing with the previous example. Both results follow from the same σ^2/n bound; $L^2 \Rightarrow$ in prob makes the WLLN automatic once we establish L^2 convergence. But the modes are genuinely distinct:

- L^2 gives *quantitative control*: $\mathbb{E}(\bar{X}_n - \mu)^2 = \sigma^2/n$ tells us the rate.
- In probability only says large deviations become unlikely; it says nothing about the expected squared error.

Concrete gap. If X_i are i.i.d. Cauchy, then $\bar{X}_n \overset{d}{=} X_1$ for every n (stable distributions), so $\bar{X}_n \not\xrightarrow{\mathbb{P}} 0$ and L^2 convergence fails ($\mathbb{E}X_1^2 = \infty$). The L^2 method requires finite variance; the Cauchy case falls outside its scope.

Convergence in distribution

Definition

X_n converges to X **in distribution** ($X_n \xrightarrow{d} X$) if

$$F_n(x) \longrightarrow F(x)$$

at every continuity point x of $F(x) = \mathbb{P}(X \leq x)$, where $F_n(x) = \mathbb{P}(X_n \leq x)$.

Key distinctions.

- Only the *distributions* (CDFs) need converge, not the random variables themselves.
- X_n and X need not live on the same probability space.
- The limit X is determined only up to equality in distribution.

Note. This is the weakest mode. The Central Limit Theorem ($\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} N(0,1)$) is a statement about convergence in distribution. We return to this in Lecture 5.

Example: convergence in distribution — Poisson approximation

Let $X_n \sim \mathrm{Binomial}(n, \lambda/n)$ for fixed $\lambda > 0$. Let $X \sim \mathrm{Poisson}(\lambda)$.

Claim: $X_n \xrightarrow{d} X$.

Proof.

For each fixed $k = 0, 1, 2, \ldots$,

$$\mathbb{P}(X_n = k) = \binom{n}{k}\left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}.$$

As $n \to \infty$: $\binom{n}{k}/n^k \to 1/k!$, $(1 - \lambda/n)^n \to e^{-\lambda}$, and $(1 - \lambda/n)^{-k} \to 1$. So

$$\mathbb{P}(X_n = k) \longrightarrow \frac{\lambda^k e^{-\lambda}}{k!} = \mathbb{P}(X = k).$$

Since F jumps only at integers, the continuity points of F are the non-integers; hence $F_n(x) \to F(x)$ at every continuity point. □

What this illustrates. There is *no single probability space* on which all X_n and X naturally live together — each X_n is defined separately. Convergence in distribution only cares that the *laws* converge. This is genuinely weaker: we cannot ask "does $|X_n(\omega) - X(\omega)| \to 0$?" because X_n and X may live on entirely different spaces.

Two indispensable tools: Markov and Chebyshev

We will use these inequalities constantly.

Proposition (Markov's inequality)

For any $X \geq 0$ and $a > 0$: $\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$.

Proof.

$\mathbb{E}[X] \geq \mathbb{E}[X\mathbf{1}_{\{X \geq a\}}] \geq a\,\mathbb{P}(X \geq a)$. □

Proposition (Chebyshev's inequality)

For any X with $\mathbb{E}[X^2] < \infty$ and $a > 0$: $\mathbb{P}(|X - \mathbb{E}X| \geq a) \leq \mathrm{Var}(X)/a^2$.

Proof.

Apply Markov to $|X - \mathbb{E}X|^2$ with threshold a^2. □

More generally, Markov applied to $|X|^p$ gives $\mathbb{P}(|X| \geq a) \leq a^{-p}\mathbb{E}|X|^p$. This is the direct link between L^p convergence and convergence in probability: if $\mathbb{E}|X_n - X|^p \to 0$, then for each $\varepsilon > 0$, $\mathbb{P}(|X_n - X| \geq \varepsilon) \leq \varepsilon^{-p}\mathbb{E}|X_n - X|^p \to 0$.

The hierarchy of implications

The four modes sit in a strict hierarchy:

$$L^p \text{ convergence } (p \geq 1) \quad \Rightarrow \quad \text{in probability} \quad \Rightarrow \quad \text{in distribution}$$

$$\text{a.s. convergence} \quad \Rightarrow \quad \text{in probability}$$

None of the reverse implications hold in general.

One important partial converse: in probability $\Rightarrow$ a.s. *along a subsequence*.

Plan:

- Prove the two forward implications.
- Give explicit counterexamples showing the gaps.
- Prove the subsequence theorem.

Proof: a.s. $\Rightarrow$ in probability

Proposition

If $X_n \xrightarrow{\text{a.s.}} X$ *then* $X_n \xrightarrow{\mathbb{P}} X$.

Proof.

Fix $\varepsilon > 0$. Define $A_n = \left\{\sup_{k \geq n} |X_k - X| > \varepsilon\right\}$.
Note that $A_n \supseteq A_{n+1}$ (the sup is over a smaller set of indices as n grows), so $A_n \downarrow$. By continuity of probability for decreasing sets:

$$\lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right).$$

The intersection equals $\{|X_n - X| > \varepsilon \text{ infinitely often}\} \subseteq \{X_n \not\to X\}$.
Since $X_n \xrightarrow{\text{a.s.}} X$, we have $\mathbb{P}(X_n \not\to X) = 0$, so $\mathbb{P}(A_n) \to 0$.
Finally, $\{|X_n - X| > \varepsilon\} \subseteq A_n$, so $\mathbb{P}(|X_n - X| > \varepsilon) \leq \mathbb{P}(A_n) \to 0$. $\square$

Proof: $L^p \Rightarrow$ in probability

Proposition

If $X_n \xrightarrow{L^p} X$ for some $p \geq 1$, then $X_n \xrightarrow{\mathbb{P}} X$.

Proof.

Fix $\varepsilon > 0$. Apply Markov's inequality to the non-negative r.v. $|X_n - X|^p$:

$$\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{P}(|X_n - X|^p > \varepsilon^p) \leq \frac{\mathbb{E}|X_n - X|^p}{\varepsilon^p} \to 0.$$

□

This is the workhorse of the weak laws.

The L^2 strategy for proving $S_n/n \xrightarrow{\mathbb{P}} \mu$ is:

1. Compute $\mathbb{E}\left[(S_n/n - \mu)^2\right] = \mathrm{Var}(S_n)/n^2$.
2. Show this $\to 0$ using variance bounds.
3. Conclude convergence in probability by the proposition above.

We execute this in Lecture 2. The point is that *computing variances is enough*.

Counterexample: in probability $\not\Rightarrow$ a.s. — the typewriter sequence

Let $(\Omega, \mathcal{F}, \mathbb{P}) = \big((0,1), \mathcal{B}(0,1), \text{Lebesgue}\big)$.
Construction. Write each $n \geq 1$ uniquely as $n = 2^k + j$ with $0 \leq j < 2^k$. Define

$$X_n(\omega) = \mathbf{1}_{\left[j \cdot 2^{-k},\, (j+1) \cdot 2^{-k}\right)}(\omega).$$

The first few: $X_1 = \mathbf{1}_{[0,1)}$, $X_2 = \mathbf{1}_{[0,1/2)}$, $X_3 = \mathbf{1}_{[1/2,1)}$, $X_4 = \mathbf{1}_{[0,1/4)}$, $X_5 = \mathbf{1}_{[1/4,1/2)}$, $\ldots$
A window of width 2^{-k} sweeps across $(0,1)$ in 2^k steps, then halves and sweeps again.

Convergence in probability to 0: $\mathbb{P}(|X_n| > 0) = \mathbb{P}(X_n = 1) = 2^{-k} \to 0$ as $n \to \infty$.

Failure of a.s. convergence: For every $\omega \in (0,1)$, the window passes over ω at every resolution k, so $X_n(\omega) = 1$ infinitely often. Thus $X_n(\omega) \not\to 0$ for any ω.

$$X_n \xrightarrow{\mathbb{P}} 0 \qquad \text{but} \qquad X_n \not\xrightarrow{\text{a.s.}} 0.$$

Counterexample: a.s. $\not\Rightarrow L^1$ — escaping mass

Construction. Let $\Omega = (0,1)$ with Lebesgue measure. Define

$$X_n(\omega) = n \cdot \mathbf{1}_{(0,1/n)}(\omega).$$

Almost sure convergence: For any fixed $\omega \in (0,1)$, once $n > 1/\omega$ we have $\omega \notin (0,1/n)$, so $X_n(\omega) = 0$. Thus $X_n(\omega) \to 0$ for *every* $\omega \in (0,1)$: this is in fact pointwise (not just a.s.) convergence.

Failure of L^1 convergence:

$$\mathbb{E}[X_n] = n \cdot \mathbb{P}\big((0,1/n)\big) = n \cdot \frac{1}{n} = 1 \not\to 0.$$

So $\mathbb{E}|X_n - 0| = 1 \not\to 0$ and $X_n \overset{L^1}{\not\longrightarrow} 0$.

Lesson. The function values concentrate near 0 for each fixed ω, but the *mass* accumulates on a shrinking set $(0,1/n)$ where the function is large, preventing L^1 convergence. This phenomenon is controlled by *uniform integrability*.

Summary: what fails and why

Implication that fails	**Counterexample**
In prob $\not\Rightarrow$ a.s.	Typewriter: $\mathbf{1}_{[j2^{-k},(j+1)2^{-k})}$, bad set moves
A.s. $\not\Rightarrow L^1$	$n \cdot \mathbf{1}_{(0,1/n)}$: pointwise $\to 0$ but $\mathbb{E}X_n = 1$
$L^p \not\Rightarrow$ a.s.	Typewriter also works: $\mathbb{E}X_n = 2^{-k} \to 0$
In prob $\not\Rightarrow L^p$	$n^{1/p} \cdot \mathbf{1}_{[j2^{-k},(j+1)2^{-k})}$: in prob $\to 0$, $\mathbb{E}\lvert X_n\rvert^p = 1$

The root cause. In probability and a.s. diverge because the "bad set" can move. L^p and a.s. diverge because mass can escape without affecting pointwise limits.

In probability $\Rightarrow$ a.s. along a subsequence

Theorem

$X_n \xrightarrow{\mathbb{P}} X$ if and only if every subsequence $X_{n(m)}$ has a further subsequence $X_{n(m_k)}$ with $X_{n(m_k)} \xrightarrow{\text{a.s.}} X$.

Proof.

($\Rightarrow$) Let $X_{n(m)}$ be any subsequence. Since $X_{n(m)} \xrightarrow{\mathbb{P}} X$, for each $k \geq 1$ we can choose $m_k > m_{k-1}$ with

$$\mathbb{P}\big(|X_{n(m_k)} - X| > 2^{-k}\big) \leq 2^{-k}.$$

Then $\sum_{k=1}^{\infty} \mathbb{P}(|X_{n(m_k)} - X| > 2^{-k}) \leq \sum 2^{-k} < \infty$. By BC1 (proved shortly): $|X_{n(m_k)} - X| > 2^{-k}$ for only finitely many k a.s., so $X_{n(m_k)} \xrightarrow{\text{a.s.}} X$.

($\Leftarrow$) If $X_n \overset{\mathbb{P}}{\not\to} X$, there exist $\varepsilon, \delta > 0$ and a subsequence $n(m)$ with $\mathbb{P}(|X_{n(m)} - X| > \varepsilon) \geq \delta$ for all m. No further subsequence can converge a.s. to X. $\square$

Continuous mapping in probability

Proposition

If $f : \mathbb{R} \to \mathbb{R}$ is continuous and $X_n \xrightarrow{\mathbb{P}} X$, then $f(X_n) \xrightarrow{\mathbb{P}} f(X)$. If additionally f is bounded, then $\mathbb{E}f(X_n) \to \mathbb{E}f(X)$.

Proof.

Given any subsequence $f(X_{n(m)})$, the theorem above gives a further sub-subsequence along which $X_{n(m_k)} \xrightarrow{\text{a.s.}} X$. Since f is continuous, $f(X_{n(m_k)}) \xrightarrow{\text{a.s.}} f(X)$, so in particular $f(X_{n(m_k)}) \xrightarrow{\mathbb{P}} f(X)$. By the subsequence theorem (in the $\Leftarrow$ direction), $f(X_n) \xrightarrow{\mathbb{P}} f(X)$.

For bounded f: along any sub-subsequence where $X_{n(m_k)} \xrightarrow{\text{a.s.}} X$, the bounded convergence theorem gives $\mathbb{E}f(X_{n(m_k)}) \to \mathbb{E}f(X)$. The same subsequence argument yields $\mathbb{E}f(X_n) \to \mathbb{E}f(X)$. $\square$

Application. If $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$ and g is continuous, then $g(\bar{X}_n) \xrightarrow{\mathbb{P}} g(\mu)$. This is the *delta method* precursor: the exact continuous mapping theorem in probability.

lim sup and lim inf of events: definitions

Let $A_1, A_2, \ldots \in \mathcal{F}$.

Definition

$$\limsup_{n\to\infty} A_n \;=\; \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k \;=\; \{\omega : \omega \in A_n \text{ for infinitely many } n\}.$$

$$\liminf_{n\to\infty} A_n \;=\; \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k \;=\; \{\omega : \omega \in A_n \text{ for all but finitely many } n\}.$$

We write $\{A_n \text{ i.o.}\} := \limsup_n A_n$ ("A_n infinitely often").

Basic facts. $\liminf A_n \subseteq \limsup A_n$, and $\mathbb{P}(\limsup A_n) \geq \limsup \mathbb{P}(A_n)$.

The fundamental connection.

$$X_n \xrightarrow{\text{a.s.}} X \iff \mathbb{P}\big(|X_n - X| > \varepsilon \text{ i.o.}\big) = 0 \quad \text{for every } \varepsilon > 0.$$

So to prove almost sure convergence, we need to control events of the form $\{A_n \text{ i.o.}\}$.

Examples of lim sup events

Example 1. $A_n = \{|X_n - X| > \varepsilon\}$. Then $\{A_n \text{ i.o.}\}$ is the event that X_n deviates from X by more than ε infinitely often. $X_n \xrightarrow{\text{a.s.}} X$ iff $\mathbb{P}(A_n \text{ i.o.}) = 0$ for all $\varepsilon > 0$.

Example 2. $X_1, X_2, \ldots$ i.i.d. Poisson(1), $A_n = \{X_n = n\}$. Then $\mathbb{P}(A_n) = e^{-1}/n!$ and $\sum_n \mathbb{P}(A_n) < \infty$. So $\mathbb{P}(A_n \text{ i.o.}) = 0$ by BC1.

Example 3. $X_1, X_2, \ldots$ i.i.d. standard exponential, $A_n = \{X_n \geq \log n\}$. Then $\mathbb{P}(A_n) = e^{-\log n} = 1/n$ and $\sum_n 1/n = \infty$. Since A_n are independent, $\mathbb{P}(A_n \text{ i.o.}) = 1$ by BC2.

Example 4. $X_1, X_2, \ldots$ i.i.d. with $\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = 1/2$, $A_n = \{X_n = 1\}$. Then $\mathbb{P}(A_n) = 1/2$, $\sum 1/2 = \infty$, and A_n independent, so $\mathbb{P}(A_n \text{ i.o.}) = 1$: we see infinitely many heads.

Borel–Cantelli Lemma 1

Theorem (First Borel–Cantelli Lemma)

Let $A_1, A_2, \ldots \in \mathcal{F}$. *If* $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, *then* $\mathbb{P}(A_n \text{ i.o.}) = 0$.

Proof.

Let $N(\omega) = \sum_{n=1}^{\infty} \mathbf{1}_{A_n}(\omega)$ count the number of events containing ω.

By linearity of expectation and the monotone convergence theorem:

$$\mathbb{E}[N] = \sum_{n=1}^{\infty} \mathbb{E}[\mathbf{1}_{A_n}] = \sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty.$$

A random variable with finite expectation is finite a.s., so $N < \infty$ a.s.

Since $N(\omega) = \infty$ precisely when $\omega \in A_n$ for infinitely many n:

$$\mathbb{P}(A_n \text{ i.o.}) = \mathbb{P}(N = \infty) = 0. \qquad \square$$

BC1 as an a.s. convergence criterion

Proposition

If $\sum_{n=1}^{\infty} \mathbb{P}(|X_n - X| > \varepsilon) < \infty$ *for every* $\varepsilon > 0$, *then* $X_n \xrightarrow{\text{a.s.}} X$.

Proof.

Fix $\varepsilon > 0$ and let $A_n^\varepsilon = \{|X_n - X| > \varepsilon\}$. The hypothesis gives $\sum_n \mathbb{P}(A_n^\varepsilon) < \infty$, so by BC1, $\mathbb{P}(A_n^\varepsilon \text{ i.o.}) = 0$.
This means: for a.e. ω, there exists $N(\omega, \varepsilon)$ such that $|X_n(\omega) - X(\omega)| \leq \varepsilon$ for all $n \geq N(\omega, \varepsilon)$.
Taking $\varepsilon = 1/m$ for $m = 1, 2, \ldots$ and unioning over (countably many) null sets:
$\mathbb{P}(\bigcup_{m=1}^{\infty} \{|X_n - X| > 1/m \text{ i.o.}\}) = 0$. On the complement (full probability),
$X_n(\omega) \to X(\omega)$. $\square$

Standard strategy. To prove $X_n \xrightarrow{\text{a.s.}} 0$: find $f(n, \varepsilon)$ with $\mathbb{P}(|X_n| > \varepsilon) \leq f(n, \varepsilon)$ and $\sum_n f(n, \varepsilon) < \infty$.

Borel–Cantelli Lemma 2

Theorem (Second Borel–Cantelli Lemma)

Let $A_1, A_2, \ldots$ be independent events with $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$. Then $\mathbb{P}(A_n \text{ i.o.}) = 1$.

Proof.

Fix M. It suffices to show $\mathbb{P}(\bigcup_{n=M}^{\infty} A_n) = 1$, since these events decrease to $\{A_n \text{ i.o.}\}$ as $M \to \infty$.
For any $M \leq N$, independence and $1 - x \leq e^{-x}$ give:

$$\mathbb{P}\left(\bigcap_{n=M}^{N} A_n^c\right) = \prod_{n=M}^{N} (1 - \mathbb{P}(A_n)) \leq \exp\left(-\sum_{n=M}^{N} \mathbb{P}(A_n)\right) \to 0$$

as $N \to \infty$, since $\sum_{n=M}^{\infty} \mathbb{P}(A_n) = \infty$.
So $\mathbb{P}(\bigcap_{n=M}^{\infty} A_n^c) = 0$, giving $\mathbb{P}(\bigcup_{n=M}^{\infty} A_n) = 1$ for every M.
Since $\bigcup_{n=M}^{\infty} A_n \downarrow \{A_n \text{ i.o.}\}$, continuity from above gives $\mathbb{P}(A_n \text{ i.o.}) = 1$. $\square$

Why independence is necessary in BC2

Remark

Independence is *essential* in the second lemma.

Counterexample. Let $A_n = A$ for all n, where $0 < \mathbb{P}(A) < 1$. Then $\sum_n \mathbb{P}(A_n) = \infty$, but $\{A_n \text{ i.o.}\} = A$ and $\mathbb{P}(A_n \text{ i.o.}) = \mathbb{P}(A) \neq 1$.

A 0-1 law for independent events. Combining BC1 and BC2: if $A_1, A_2, \ldots$ are independent, then

$$\mathbb{P}(A_n \text{ i.o.}) = \begin{cases} 0 & \text{if } \sum_n \mathbb{P}(A_n) < \infty, \\ 1 & \text{if } \sum_n \mathbb{P}(A_n) = \infty. \end{cases}$$

The boundary is decided entirely by the sum of the probabilities.

This is a special instance of **Kolmogorov's 0-1 law**: any tail event of an independent sequence has probability 0 or 1. The event $\{A_n \text{ i.o.}\}$ is a tail event (it is not affected by changing finitely many of the A_n).

Necessity of $\mathbb{E}|X| < \infty$ for the strong law

Theorem

Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}|X_1| = \infty$. Then

$$\mathbb{P}(|X_n| \geq n \text{ i.o.}) = 1.$$

Consequently, $\mathbb{P}(\lim_{n\to\infty} S_n/n$ exists and is finite$) = 0$.

Proof.

Step 1. We claim $\sum_n \mathbb{P}(|X_1| \geq n) = \infty$. From the layer-cake identity (proved carefully next lecture): $\mathbb{E}|X_1| = \int_0^\infty \mathbb{P}(|X_1| > t)\, dt \leq \sum_{n=0}^\infty \mathbb{P}(|X_1| > n)$. So $\mathbb{E}|X_1| = \infty$ forces the sum to diverge. Since the events $\{|X_n| \geq n\}$ are independent (i.i.d. sequence), BC2 applies.

Step 2. On the set $C = \{\lim S_n/n = L \in \mathbb{R}\}$, we have $S_{n+1}/n - S_n/n = X_{n+1}/n \to 0$, which forces $|X_n|/n \to 0$ a.s. on C. But $|X_n| \geq n$ i.o. (from Step 1) means $|X_n|/n \geq 1$ i.o., a contradiction. So $\mathbb{P}(C) = 0$. $\square$

A first strong law: the fourth-moment condition

Theorem

Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}X_i = \mu$ and $\mathbb{E}X_i^4 < \infty$. Let $S_n = X_1 + \cdots + X_n$. Then

$$\frac{S_n}{n} \xrightarrow{\text{a.s.}} \mu.$$

WLOG $\mu = 0$ (replace X_i by $X_i - \mu$). The strategy is:

1. Bound $\mathbb{E}[S_n^4]$ using independence and the moment condition.
2. Apply Chebyshev with the fourth moment to get a rate on $\mathbb{P}(|S_n/n| > \varepsilon)$.
3. Sum and apply BC1.

Remark. The condition $\mathbb{E}X^4 < \infty$ is far stronger than necessary. The general strong law (Lecture 3) requires only $\mathbb{E}|X| < \infty$. This proof is given as a clean illustration of the Chebyshev + Borel–Cantelli method.

Step 1: computing $\mathbb{E}[S_n^4]$

Expand $S_n^4 = \left(\sum_{i=1}^n X_i\right)^4 = \sum_{i,j,k,\ell=1}^n X_i X_j X_k X_\ell$.
By independence and $\mathbb{E}X_i = 0$: $\mathbb{E}[X_i X_j X_k X_\ell] = 0$ whenever any index appears exactly once (factor out that X; it is independent of the rest with mean 0). So only terms where every index appears at least twice contribute.

Case 1: all four indices equal ($i = j = k = \ell$). n such terms, each contributing $\mathbb{E}X_1^4$.
Case 2: exactly two distinct values, each appearing exactly twice. Choose 2 distinct indices: $\binom{n}{2}$ ways. Choose which of the 3 pairings $\{\{ij, k\ell\}, \{ik, j\ell\}, \{i\ell, jk\}\}$ to use: 3 ways. Total: $3n(n-1)$ terms. Each contributes $(\mathbb{E}X_1^2)^2$ by independence.

Therefore:

$$\mathbb{E}[S_n^4] = n\,\mathbb{E}X_1^4 + 3n(n-1)(\mathbb{E}X_1^2)^2 \leq Cn^2,$$

where $C = \mathbb{E}X_1^4 + 3(\mathbb{E}X_1^2)^2 < \infty$.

Steps 2 and 3: Chebyshev and Borel–Cantelli

Step 2: Chebyshev with the fourth moment.

For any $\varepsilon > 0$, apply Markov to S_n^4 with threshold $(n\varepsilon)^4$:

$$\mathbb{P}\left(\left|\frac{S_n}{n}\right| > \varepsilon\right) = \mathbb{P}(|S_n|^4 > n^4\varepsilon^4) \leq \frac{\mathbb{E}[S_n^4]}{n^4\varepsilon^4} \leq \frac{Cn^2}{n^4\varepsilon^4} = \frac{C}{n^2\varepsilon^4}.$$

Step 3: Apply BC1.

Since $\sum_{n=1}^{\infty} \frac{C}{n^2\varepsilon^4} < \infty$, the first Borel–Cantelli lemma gives

$$\mathbb{P}\left(\left|\frac{S_n}{n}\right| > \varepsilon \text{ i.o.}\right) = 0.$$

Since $\varepsilon > 0$ is arbitrary, we conclude $\frac{S_n}{n} \xrightarrow{\text{a.s.}} 0 = \mu$. $\square$

Observation. The key algebraic fact was $\mathbb{E}S_n^4 = O(n^2)$. Dividing by n^4 produces a summable $O(n^{-2})$ bound. This is why the fourth moment suffices.

Summary

We proved today:

- Four modes: a.s., in probability, L^p, in distribution; and the implications $L^p \Rightarrow$ in prob, a.s. $\Rightarrow$ in prob.
- Explicit counterexamples (typewriter, $n \cdot \mathbf{1}_{(0,1/n)}$) showing the gaps are real.
- Subsequence theorem: in prob $\Rightarrow$ a.s. along a subsequence.
- **BC1**: $\sum \mathbb{P}(A_n) < \infty \Rightarrow \mathbb{P}(A_n \text{ i.o.}) = 0$.
- **BC2**: independence $+ \sum \mathbb{P}(A_n) = \infty \Rightarrow \mathbb{P}(A_n \text{ i.o.}) = 1$.
- Necessity of $\mathbb{E}|X| < \infty$ (via BC2) and first SLLN under $\mathbb{E}X^4 < \infty$ (via BC1).